Exposé VII

COHOMOLOGY OF SOME CLASSICAL SCHEMES
AND
THE COHOMOLOGICAL THEORY OF CHERN CLASSES

by

J.-P. Jouanolou

Throughout the text, an integer $\nu > 0$ is fixed and we put

$$A = \mathbb{Z}/\nu\mathbb{Z};$$

the residual characteristics of all schemes considered are assumed to be prime to the integer ν. If X is such a scheme, μ_X denotes the sheaf of ν-th roots of unity and $A^\bullet(X)$ denotes the cohomology ring

$$\bigoplus_i H^{2i}(X, \mu_X^{\otimes i}),$$

with multiplication given by cup-product.

1. Vector bundles

Let S be a scheme and let E be a locally free $\mathcal{O}_S$-module, understood to be of finite type. Put $X = \mathrm{V}(E)$ and $U = \mathrm{V}(E)^*$, the complement of the zero section, and denote by

$$p : X \to S, \qquad q : U \to S$$

the canonical projections.

Proposition 1.1. Let L be an A-module.

(i) a) For every integer $j > 0$, one has

$$R^j p_*(p^* L) = 0.$$

b) The adjunction morphism

$$L \longrightarrow p_* p^*(L)$$

is an isomorphism.

(ii) Suppose that E has constant rank r.

a)

$$R^i_! p(p^* L) = 0 \quad \text{if } i \neq 2r.$$

b) The trace morphism (SGA 4 XVIII)

$$R^{2r}_! p(p^* L) \longrightarrow L \otimes \mu_S^{\otimes -r}$$

is an isomorphism.

Proof. Part (i) follows immediately by localization for the Zariski topology from (SGA 4 XV 2.2). Assertion (ii) is essentially contained in (SGA 4 XVIII).

Corollary 1.2. Let L be an A_S-module. For every integer i, the inverse-image morphism

$$H^i(S, L) \longrightarrow H^i(X, p^*L)$$

is an isomorphism.

Proof. By (1.1), the adjunction morphism $L \to Rp_*(p^*L)$ is an isomorphism. Since $H^*(X, p^*L) = H^*(S, Rp_*p^*(L))$, the assertion follows.

Proposition 1.3. Suppose that E has constant rank r. For every A_S-module L, the following properties hold.

(i) a)

$$R^i q_*(q^*L) = 0 \quad \text{if } i \neq 0, 2r-1.$$

b) The adjunction morphism

$$L \longrightarrow q_*q^*(L)$$

is an isomorphism.

c) If Y denotes the complement of U in X, then the morphism

$$R^{2r-1}q_*(q^*L) \longrightarrow L \otimes \mu_S^{\otimes -r}$$

obtained as the composite of the boundary morphism

$$R^{2r-1}q_*(q^*L) \longrightarrow R^{2r}_Y p_*(p^*L) \tag{1}$$

and the purity isomorphism (SGA 4 XVIII)

$$R^{2r}_Y p_*(p^*L) \longrightarrow L \otimes \mu_S^{\otimes -r}$$

is an isomorphism.

(ii) a)

$$R^i_! q(q^*L) = 0 \quad \text{if } i \neq 1, 2r.$$

b) The boundary morphism

$$L \longrightarrow R^1_! q(q^*L)$$

coming from the unlimited exact sequence, where $f : Y \to S$ is the canonical projection,

$$\cdots \longrightarrow R^i_! q(q^*L) \longrightarrow R^i_! p(p^*L) \longrightarrow R^i f_*(f^*L) \longrightarrow R^{i+1}_! q(q^*L) \longrightarrow \cdots$$

is an isomorphism.

c) The trace morphism

$$R^{2r}_! q(q^*L) \longrightarrow L \otimes \mu_S^{\otimes -r}$$

is an isomorphism.

Proof. Let $\alpha : U \to X$ and $\beta : Y \to X$ be the canonical immersions. One knows (SGA 4 XVI 3.7) that, putting $F = p^*(L)$,

$$F \xrightarrow{\sim} \alpha_*\alpha^*(F) \quad \text{and} \quad (R^j\alpha_*)\alpha^*(F) = 0 \quad \text{if } j \neq 0, 2r-1.$$

Moreover, the sheaves $(R^j\alpha_*)\alpha^*F$ for $j \geq 1$ have support in Y, and hence are acyclic for the functor p_*. From these remarks one obtains assertions (i) a) and b) by using the Leray spectral sequence

$$R^ip_*(R^j\alpha_*)(q^*L) \Longrightarrow R^nq_*(q^*L).$$

Assertion (i) c) follows from (1.1 (i)) and from the unlimited exact sequence

$$\cdots \longrightarrow R^i_Yp_*(p^*L) \longrightarrow R^ip_*(p^*L) \longrightarrow R^iq_*(q^*L) \longrightarrow R^{i+1}_Yp_*(p^*L) \longrightarrow \cdots.$$

The sheaf $R^{2r}_Yp_*(p^*L)$ is associated with the presheaf

$$T \longmapsto H^{2r}_{Y\times_S T}(X \times_S T, p^*L).$$

We prove (ii). Assertion c) comes from the fact that q is smooth and has connected geometric fibres of dimension r. The exact sequence mentioned in b) is obtained by applying the functor $R_!p$ to the canonical exact sequence

$$0 \longrightarrow \alpha_!(q^*L) \longrightarrow p^*L \longrightarrow \beta_!(\beta^*p^*L) \longrightarrow 0.$$

Using (1.1 (i)), one obtains assertions (ii) a) and b) without difficulty from this unlimited exact sequence.

Remark 1.4. For $S = \mathrm{Spec}(\mathbb{C})$, (1.3) says that the complement of the origin in $\mathbb{C}^r$ has the same cohomology as the real sphere of dimension $2r-1$, a deformation retract of $\mathbb{C}^r - 0$.

Corollary 1.5 (Gysin exact sequence). The Leray spectral sequence

$$H^i(S, R^jq_*(q^*L)) \Rightarrow H^{i+j}(U, q^*L)$$

defines an unlimited exact sequence, called the Gysin exact sequence,

$$\cdots \longrightarrow H^i(S, L) \longrightarrow H^i(U, q^*L) \longrightarrow H^{i-2r+1}(S, L \otimes \mu_S^{\otimes -r}) \longrightarrow H^{i+1}(S, L) \longrightarrow \cdots.$$

Proof. Lemma of two lines.

2. Projective schemes

In what follows, the mode of exposition adopted is based on a suggestion of L. Illusie, and the method followed is close to a method of P. Deligne for proving the degeneration of certain spectral sequences [6].

2.1

Let X and Y be two schemes and let $f : X \to Y$ be a morphism. We shall define an arrow

$$Rf_*(A_X) \otimes Rf_*(A_X) \longrightarrow Rf_*(A_X) \tag{2.1.1}$$

in $D(Y)$, having the usual associativity and commutativity properties, and therefore endowing $Rf_*(A_X)$ with a structure of generalized algebra. For this, write

$$C^*(X, A_X)$$

for the Godement resolution of A_X (SGA 4 XVII). Proceeding as in (TF 6.6), one defines a morphism of resolutions

$$C^*(X, A_X) \otimes C^*(X, A_X) \longrightarrow C^*(X, A_X \otimes A_X).$$

Composing it with the morphism

$$C^*(X, A_X \otimes A_X) \longrightarrow C^*(X, A_X)$$

deduced from the multiplication of A_X, one finally obtains a morphism

$$C^*(X, A_X) \otimes C^*(X, A_X) \longrightarrow C^*(X, A_X). \tag{2.1.2}$$

Applying the functor f_*, one obtains without difficulty a morphism of complexes of A_Y-modules

$$f_*C^*(X, A_X) \otimes f_*C^*(X, A_X) \longrightarrow f_*C^*(X, A_X),$$

and hence (2.1.1), using flat resolutions of the components of the expression on the left.

2.2

Let S be a scheme and let E be a locally free $\mathcal{O}_S$-module of rank $r+1$. We denote by

$$P = P_S(E) \xrightarrow{p} S$$

the corresponding projective bundle. The canonical $\mathcal{O}_P$-module $\mathcal{O}_P(1)$ defines, as is known, by means of the Kummer exact sequence, an element

$$\xi \in H^2(P, \mu),$$

and we shall denote by η the image of ξ under the canonical morphism

$$H^2(P, \mu) \longrightarrow H^0(S, R^2p_*(\mu))$$

coming from the Leray spectral sequence

$$H^i(S, R^jp_*(\mu)) \Rightarrow H^{i+j}(P, \mu).$$

The class ξ is identified with an arrow $A_P \to \mu_P[2]$ in $D(P)$, which by transposition defines $\mu_P^{\otimes -1}[-2] \to A_P$, and hence by the usual adjunction an arrow

$$c : \mu_S^{\otimes -1}[2] \longrightarrow Rp_*(A_P),$$

coinciding with η on cohomology in dimension 2.

By tensor product, the arrow c defines, by means of (2.2.1), arrows

$$c_i : \mu_S^{\otimes -i}[-2i] \longrightarrow Rp_*(A_P),$$

for every integer $i \geq 1$, and we agree to denote by c_0 the adjunction arrow

$$A_S \longrightarrow Rp_*(A_P).$$

Thus one defines an arrow

$$\gamma = \bigoplus_{i=0}^{r} c_i : \bigoplus_{i=0}^{r} \mu_S^{\otimes -i}[-2i] \longrightarrow Rp_*(A_P)$$

of $D(S)$.

Theorem 2.2.1. The arrow γ is an isomorphism.

Once the arrow γ has been defined, (2.2.1) is equivalent to the following corollary.

Proposition 2.2.2. Under the preceding hypotheses, one has:

a)
$$R^{2i+1}p_*(A_P) = 0 \quad \text{for every } i \in \mathbb{N}.$$

b) When $\bigoplus_i R^{2i}p_*(\mu_P^{\otimes i})$ is endowed with its A_S-algebra structure deduced from cup-product, the morphism of A_S-algebras
$$A_S[T] \longrightarrow \bigoplus_i R^{2i}p_*(\mu_P^{\otimes i})$$
defined by η has kernel the ideal (T^{r+1}) and hence defines an isomorphism of A_S-algebras
$$\theta : A_S[T]/(T^{r+1}) \xrightarrow{\sim} \bigoplus_i R^{2i}p_*(\mu_P^{\otimes i}).$$

c) The morphism
$$A_S \longrightarrow R^{2r}p_*(\mu^{\otimes r})$$
obtained by composing θ with the restriction to A_S of multiplication by T^r on $A_S[T]/(T^{r+1})$ is the inverse of the trace morphism (SGA 4 XVIII)
$$R^{2r}p_*(\mu^{\otimes r}) \longrightarrow A_S;$$
in other words,
$$\mathrm{Tr}(\eta^r) = 1.$$

Proof. Since the data are stable under base change, the proper base-change theorem (SGA 4 XII 5.1) and compatibility of the trace morphism with base change (SGA 4 XVIII) allow one to reduce to checking the analogous assertion on the fibres of P. In other words, one may suppose that S is the spectrum of a separably closed field k and that p is the canonical morphism $\mathbb{P}_k^r \to k$. In this case, the proof uses the following lemma.

Lemma 2.2.3. Let k be a separably closed field and let $p : \mathbb{P}_k^r \to k$ be the canonical projection. Then:

a)

$$R^{2i+1}p_*(A_P) = 0 \quad \text{for every } i.$$

b)

$$R^{2i}p_*(A_P) \simeq A_k \quad \text{for } i \in [0,r], \qquad R^{2i}p_*(A_P) = 0 \quad \text{if } i > r.$$

The lemma, manifestly true for $r = 0$, is proved by induction on r. Suppose therefore that it is true for $r-1$ and prove it for r. The projective space $\mathbb{P}_k^{r-1}$ embeds as a closed subspace in $\mathbb{P}_k^r$ with complement the open $\mathbb{E}_k^r$, whence an exact sequence of cohomology with proper supports

$$\cdots \longrightarrow H_!^i(\mathbb{E}_k^r) \longrightarrow H^i(\mathbb{P}_k^r) \longrightarrow H^i(\mathbb{P}_k^{r-1}) \longrightarrow H_!^{i+1}(\mathbb{E}_k^r) \longrightarrow \cdots .$$

The computation of cohomology with proper supports of $\mathbb{E}_k^r$ (1.1) shows that, if $i \neq (2r, 2r-1)$, the inverse-image map

$$H^i(\mathbb{P}_k^r) \longrightarrow H^i(\mathbb{P}_k^{r-1})$$

is an isomorphism, whence the assertion in this case. If $i = 2r-1$, one similarly sees that $H^{2r-1}(\mathbb{P}_k^r)$ is contained in $H^{2r-1}(\mathbb{P}_k^{r-1})$, hence is zero by the induction hypothesis. Finally, if $i = 2r$, the relations

$$H^{2r-1}(\mathbb{P}_k^{r-1}) = H^{2r}(\mathbb{P}_k^{r-1}) = 0$$

show that

$$H_!^{2r}(\mathbb{E}_k^r) \xrightarrow{\sim} H^{2r}(\mathbb{P}_k^r),$$

whence the assertion by (1.1).

The lemma implies part a) of 2.2.1. We now prove that θ is an isomorphism. The morphism θ may be interpreted as a morphism of rings

$$A[T]/(T^{r+1}) \longrightarrow \bigoplus_i H^{2i}(P, \mu^{\otimes i}).$$